

AZURE MASTER ARCHITECT PROGRAM · JUNE 2026

# Unified Digital Citizen Services Platform.

One front door for 2.1 M Nordic citizens across DK · SE · NO

ARCHITECTURE, AI & AGENTIC SUBMISSION

# What this is — and what it isn't.

 **A target vision**  
The production-target architecture for a federated Nordic platform — end to end.

 **A working core**  
Deployed on a real Azure tenant with one command — the live demo runs on it.

 **Some blueprint**  
A few bricks are designed and registered, not yet switched on — each with a roadmap gate.

Every claim in this deck carries one honesty label:

 Live — on the tenant today

 Implemented — merged & smoke-tested

 Scripted — installer phase, idempotent

 Blueprint — designed, not yet live

 Roadmap — external dependency

UDCSP is a **production accelerator**: a target vision, a working core you can run today, and a named path for the rest.

# The problem.

**2.1 M**

citizens across DK SE NO

**47**

separate legacy portals

**28 d**

average time for a residency case

A citizen who moves from Copenhagen to Stockholm proves their identity again, re-sends the same papers, waits **28 days** for a decision, and uses a portal that may not speak their language — or may not be accessible at all.

UDCSP unifies the **identity** and the **experience**, then connects to the national systems to **share and check** information — never to replace them.

GDPR

EU AI Act

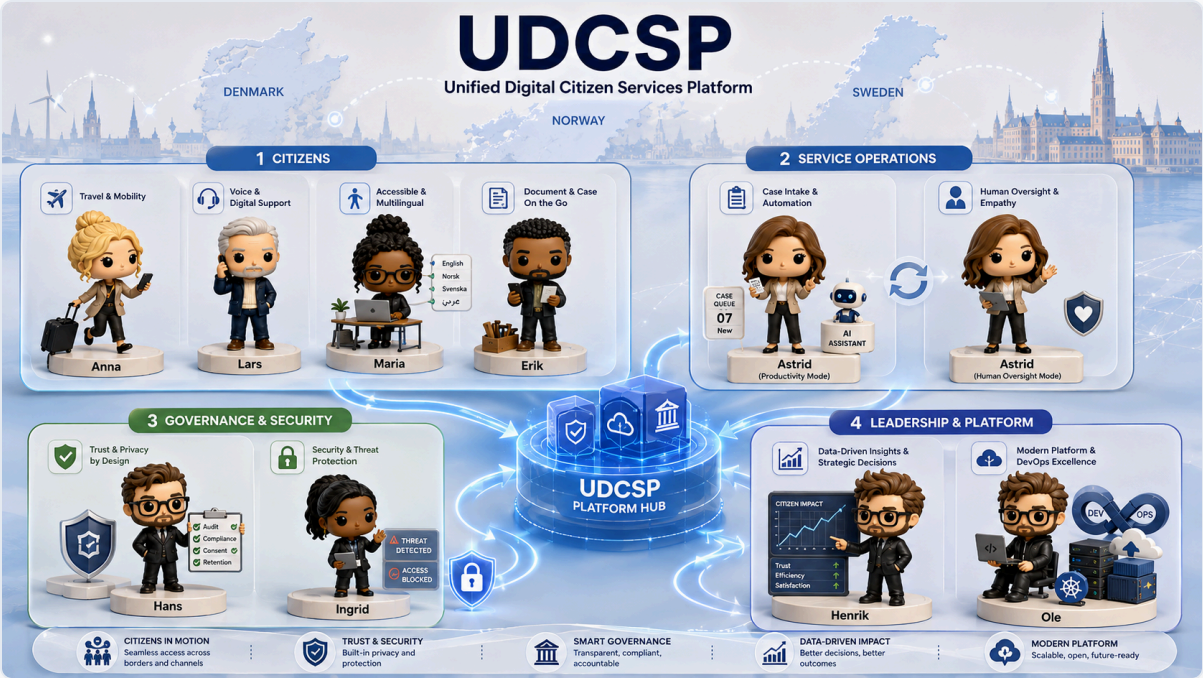
eIDAS 2.0

NIS2

WCAG 2.1 AA

ePrivacy

# Who we build for.



## CITIZENS

**Anna · Lars · Maria · Erik**

Anna moves country. Lars is blind and calls in. Maria uses a screen reader. Erik uploads from his phone.

## OPERATIONS

**Astrid — caseworker**

The AI helps her work faster. She always makes the final decision.

## GOVERNANCE

**Hans · Ingrid**

Hans protects privacy and consent. Ingrid watches for security threats.

## LEADERSHIP

**Henrik · Ole**

Henrik needs clear data to decide. Ole runs the platform with one command.



# The insight: unify the identity, connect the systems.

We did **not** rebuild the national portals. We unified the one thing that was missing — a single **identity** and a single **experience** — and connected it to the systems that already hold the records.

The citizen signs in once with their national **eID**. From there the platform **shares and checks** information with each authority: it pre-fills the right national form, submits it, and follows the status live.

borger.dk, Skatteverket, NAV, Altinn and UDI stay the system of record and the decision-maker. We **connect** to them — we never replace them.

## WHY IDENTITY-FIRST, NOT ANOTHER PORTAL

- One identity, one design, one accessibility standard
- Information is **shared and verified**, not re-collected
- The **same code** on web, mobile and phone
- A new service is a new card on the same shelf — not a new website
- The authorities keep their data; we keep the experience

Sign in once. Share and check — don't re-collect. Follow the case live.

# One front door — what the citizen lands on.

UDUDCSP Citizen Portal

HomeMy casesConsentAccessibilityComplianceDemos



EN · English

Hi Anna  Denmark citizenSign out

NORDIC PUBLIC SERVICES · LIVE

## Unified Digital Citizen Services

One trusted entry point for residents of Denmark, Sweden and Norway. Apply once, follow your case in real time, get help in your language — with caseworkers always in the loop.

Start an applicationTrack my cases

 EU data residency · sovereign by design

 12 languages · ICU MessageFormat, RTL ready

 WCAG 2.1 AA · screen-reader & keyboard verified

 EU AI Act · transparent, auditable, human override

### What can we help you with today?

Pick a service and we'll guide you. Every form pre-fills what we already know.



#### Residency transfer

One guided intake. We pre-fill the country-specific registration steps and route you to the right authority — borger.dk / CPR · Skatteverket · Skatteetaten.

[Get started →](#)



#### Tax residency certificate

Request a tax residency certificate for DK, SE or NO. We pick the right form and pre-fill it from your eID data.

[Get started →](#)



#### Child & family benefit

Check eligibility and apply for child or family benefits. Upload payslip, lease or birth certificate — we extract the data and route to the competent authority.

[Get started →](#)



#### National Tax-administration

Reach Skatteetaten, Skattestyrelsen or Skatteverket by phone. Our AI voice assistant handles refunds, deductions and residency tax in your national language — with a human caseworker one keystroke away.

[Get started →](#)

AI consent active. Manage on [Consent & privacy](#).

Citizen assistantPersonalised

Hi Anna — ask me about your applications, your case status, or how to start a new service.

Ask about your case...Send

Powered by **Foundry topic-router** via APIM. Every prompt and response is logged for audit (EU AI Act).

×

# One identity, connected to every national system.

UD

UDCSP Citizen Portal

Home

My cases

Consent

Accessibility

Compliance

Demos

EN - English

Hi Anna

Danmark citizen

Sign out

Get started →

Get started →

Get started →

Get started →

A unified platform across the Nordic public sector

UDCSP is a unified citizen platform that connects, in a single guided experience, the registers, eIDs and case systems of Denmark, Sweden and Norway. We don't replace national authorities — we collect your data once, validate the cross-border rules (Info Norden, Øresunddirekt, Grensetjänsten, EU Single Digital Gateway), pre-fill the right form, and submit it to the competent authority for your country.

UNIFIED PLATFORM

UDCSP

One guided intake

Denmark

borger.dk

Citizen portal

CPR

Population register

MitID

National eID

SKAT

Tax authority

Udbetaling DK

Family benefits

Sweden

Skatteverket

Tax & population

Försäkringskassan

Social insurance

BankID

eID

Freja+

eID

Norway

Skatteetaten

Tax & registry

NAV

Welfare & benefits

Altinn

Forms portal

UDI

Immigration

ID-porten

eID gateway

AI consent active. Manage on [Consent & privacy](#).

Citizen assistant

Personalised

Hi Anna — ask me about your applications, your case status, or how to start a new service.

Ask about your case...

Send

Powered by **Foundry topic-router** via APIM. Every prompt and response is logged for audit (EU AI Act).

×

How the platform works for you

Four simple promises behind every screen — your data stays at home, AI helps but never decides alone, the portal is usable by everyone, and you stay in charge of your information.

EU

Your data stays in your country

When you sign in as a Danish, Swedish or Norwegian resident, the portal talks only to that country's public-service systems. Nothing about you is copied or stored in another country.

AI helps, a person decides

AI can pre-read your documents, suggest the right form and give a first opinion on whether you're eligible — but a real caseworker reviews every decision and signs it off. You always see, in plain language, what the AI suggested and why.

Made for everyone

The portal is available in twelve languages, including Polish, Arabic, Sámi, Ukrainian and Finnish. You can use it entirely with a keyboard, with a screen reader, in high-contrast mode or with a dyslexia-friendly font.

You stay in charge

You decide what we may do with your data — share with another Nordic country, send you SMS reminders, forward a copy to a third party. You can change your mind at any time on the Consent page, and the change takes effect immediately.

7

# How the front door routes.

One guided intake. The portal finds **which authority owns the request** and sends it there — using each citizen's national eID and the public cross-border rules.



- **Denmark** — borger.dk · MitID



- **Sweden** — Skatteverket · BankID / Freja+



- **Norway** — UDI · Altinn · ID-porten

The rules come from Info Norden, Øresunddirekt and the EU Single Digital Gateway — public policy, not logic we invented.

The screenshot displays the UDCSP Citizen Portal interface. At the top, there's a navigation bar with links like Home, My cases, Consent, Accessibility, Compliance, and Demos. A user profile for 'Hi Anne' is visible in the top right. The main heading is 'Filing from Denmark' followed by 'Residency transfer', described as 'A single guided application that updates your civil registration with the relevant Nordic authority.' Below this, a section titled 'Connected to the official population registers' explains that UDCSP is a unified citizen platform. It lists the official eID providers for Denmark (CPR, borger.dk, MitID), Sweden (Skatteverket, BankID / Freja+, Altinn), and Norway (Skatteetaten, UDI, Altinn, ID-porten). A progress bar shows six steps: 1. Your move (active), 2. Identity (pre-filled), 3. Documents, 4. Eligibility criteria, 5. Cross-border consent, and 6. Review & submit. The 'YOUR MOVE' section prompts the user to provide their destination country (set to Norway), planned move date (05/29/2026), and destination address (e.g., Svanavägen 24, 113 57 Stockholm). A dropdown for 'DEPENDENTS MOVING WITH YOU' is set to 2.

# What we ship.

CHANNEL



Web · Mobile · Voice

One web app · 12 languages · accessibility-tested · one free phone number per country

AI BRAIN



One brain · 7 agents

One orchestrator sends work to six specialists on Azure AI Foundry

OUTCOME



28 days → 4

Cross-border residency in four days · 47 portals become one · 12 languages

SOVEREIGN



3 zones

One AI hub and one data zone per country · no data crosses a border uninvited

UDCSP is a **unified citizen platform**, not a replacement. The national authorities still make the decision.

# Sign in once — the rest is pre-filled.

UDUDCSPCitizen Portal

HomeMy casesConsentAccessibilityComplianceDemos

🇩🇰🇸🇪🇩🇰

EN · English

Hi Anna

Danmark citizen

Sign out

1Your move

2Identity (pre-filled)

3Documents

4Eligibility criteria

5Cross-border consent

6Review & submit

DOCUMENTS

Drop your employment contract — the **Document Extractor** agent (Foundry + AI Document Intelligence) reads employer, role, salary and start date. The file is stored in your country's lake (DK → udcspdkprod1ake) and never leaves the sovereign zone.

Upload employment contract (PDF / image)

employment\_contract\_anna\_jensen.pdf · 5 KB

Preview

Open in new tab

EMPLOYMENT CONTRACT

Fictional sample document generated for demonstration purposes

This Employment Contract is made on **14 May 2026** between the Employer and the Employee named t  
personal details, company details, dates, addresses, and compensation figures are fictional.

| EMPLOYER                                  | EMPLOYEE               |
|-------------------------------------------|------------------------|
| Company Name: Northbridge Consulting Ltd. | Full Name: Anna Jensen |

Extracted from your document — please confirm:

- **employeeName:** Anna Jensen
- **employer:** Nordic Retail A/S
- **employerCvr:** 12345678
- **period:** 2025-10
- **monthlyIncomeGross:** 36000

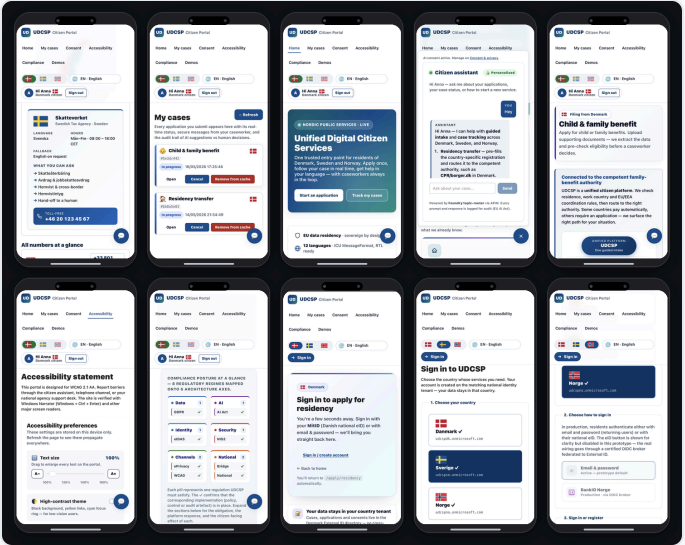
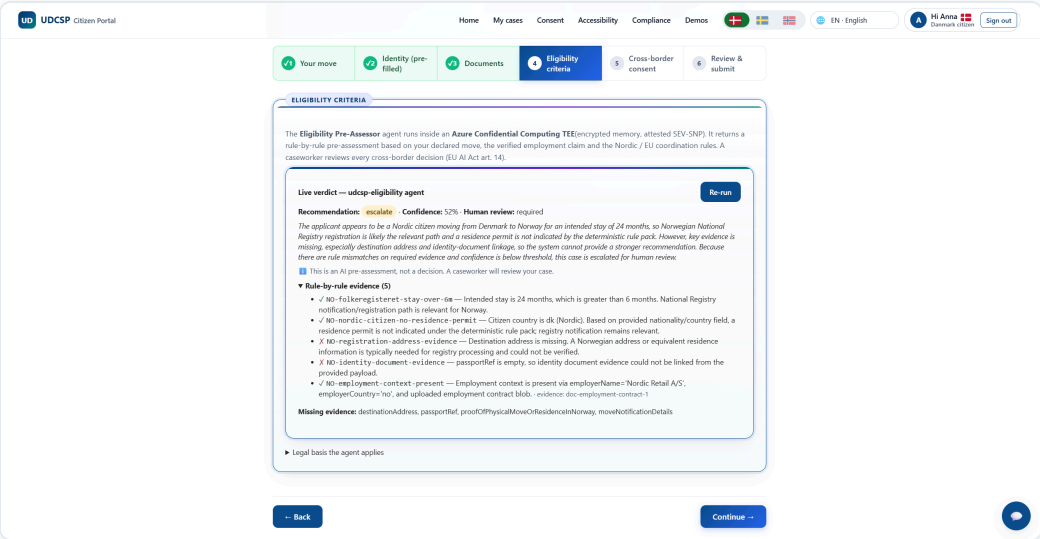
# The platform you actually see.

One web app · 12 languages · 3 channels · always accessible.

The same React and TypeScript code runs on phone, tablet and laptop, and adapts to each screen.

The chat assistant stays in the corner. The accessibility menu offers slow speech, high contrast and reduced motion.

On every *Apply* page, the citizen sees the AI result — confidence, evidence, missing documents — **before** they consent.



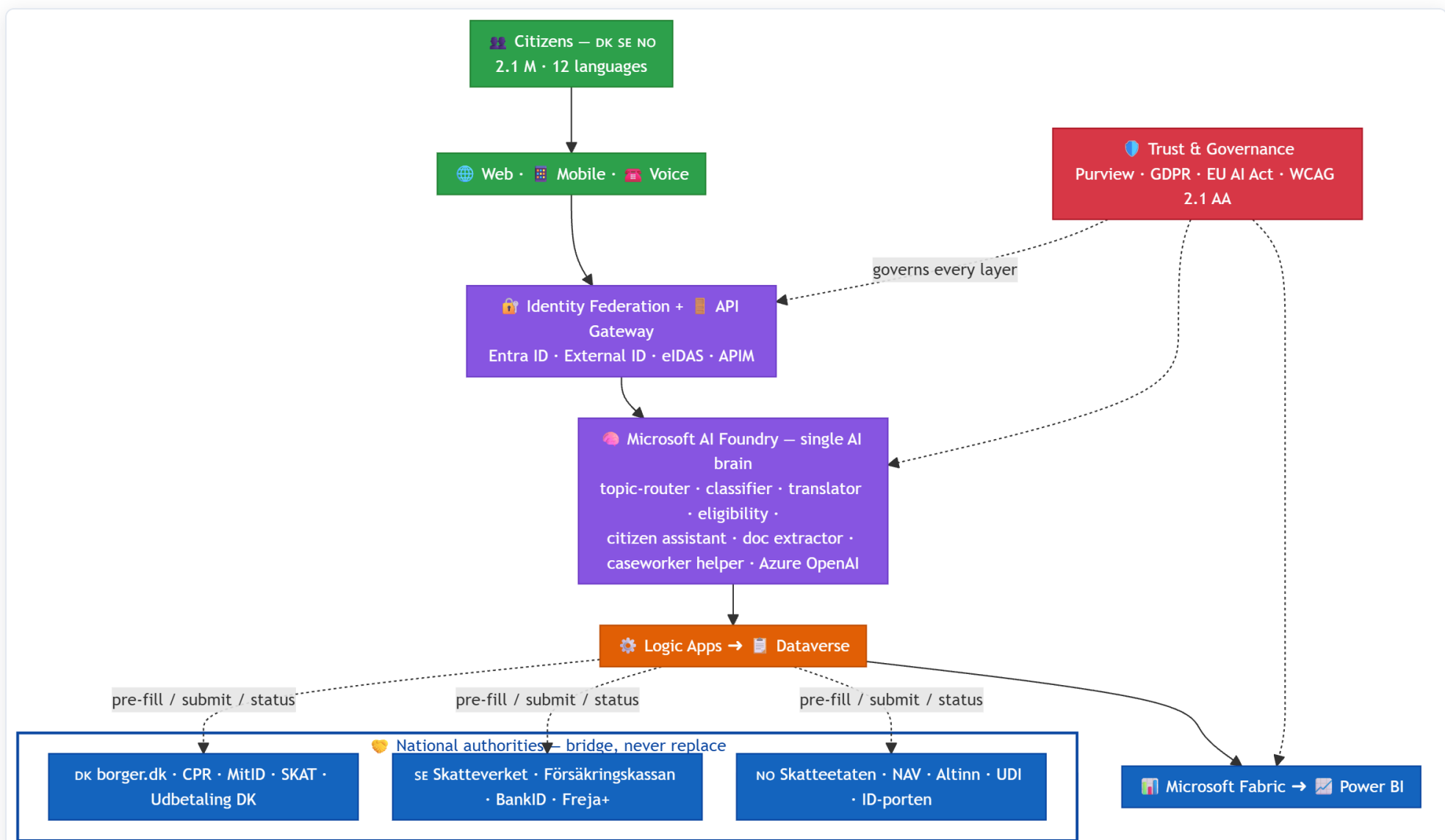
# Architecture.

How we built it — three sovereign zones, one private path from the portal to the case

01



# High-level architecture.



# One country, one hub — three times.

## A HUB PER COUNTRY

Each country has its own Azure region, private network, identity system, and **AI hub** on Azure AI Foundry.



- **Denmark** — North Europe region



- **Sweden** — Sweden Central region



- **Norway** — Norway East region

No AI call ever crosses a border.

## THE RIGHT MODEL FOR EACH JOB

- **Real-time speech** model — the voice channel
- **Strong reasoning** model — the hard decisions
- **Fast** model — simple routing and sorting

The model name comes from one parameter file, so a region change is a one-line edit.

**One honest exception.** Real-time speech is not available yet in Norway, so Norwegian calls use the Sweden hub, inside the EU Data Boundary. The audio still stays in Norway. When the model arrives, one parameter brings it home.

# Privatising every zone — from the portal to the case.

**One public door per country.** Everything behind it is private — from the citizen's first click to the caseworker's queue.

## 01 One public front door

Azure Front Door and a web firewall. Nothing else is public.

## 02 A private API gateway

Azure API Management checks the token — the only way in.

## 03 Backends on private endpoints

Workflows, agents, storage, database, vault: off the internet.

Private DNS per country.

## 04 The case lands in Dataverse

One case environment per country, in-region — Power Apps over Dataverse today, D365 Customer Service when licences land. Writes go through workflows, reads through the gateway.

## 05 Analytics copies only numbers

Dataverse sends metrics to Microsoft Fabric. The raw rows stay in the country.

## THE ONLY OTHER PUBLIC ADDRESS

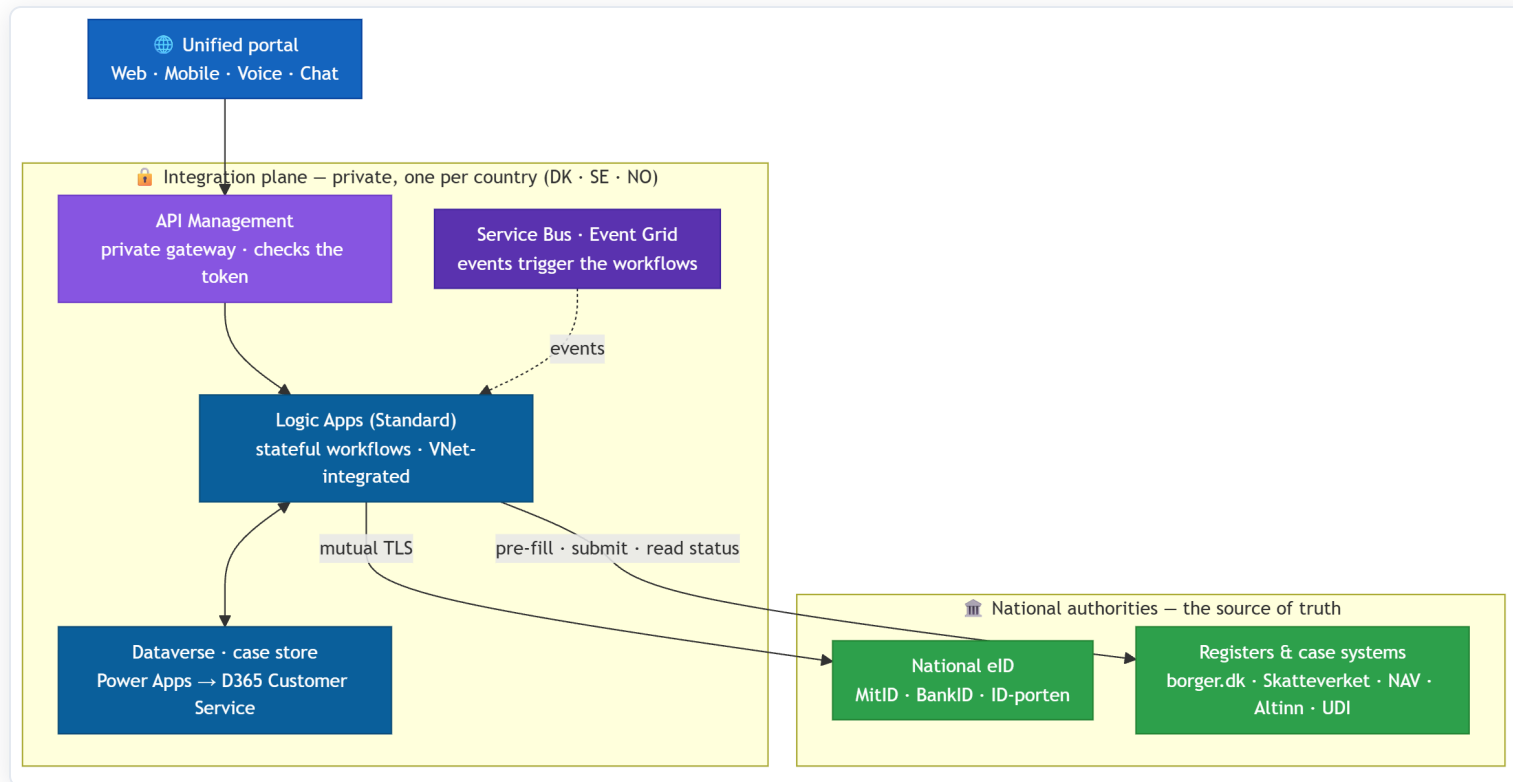
Azure Bastion — the single admin door per country. All outbound traffic goes through Azure Firewall, with allow-lists and TLS inspection.

## ENCRYPTED EVERYWHERE

- Customer-managed keys, one set per country
- Two-way TLS to the national authorities
- Field-level encryption for national ID numbers

# The integration bridge.

**One private integration plane per country.** The portal never calls a ministry directly — Logic Apps do, on its behalf.

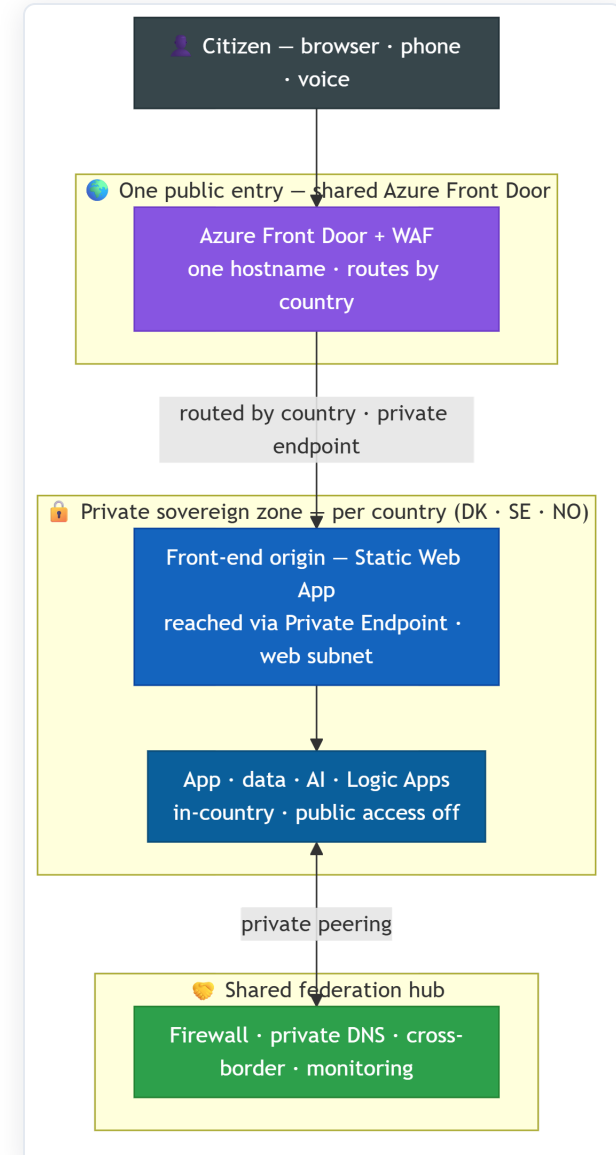


Logic Apps are the **bridge** — stateful workflows that pre-fill the national form, submit it, and read the status back, writing each case into **Dataverse** and reaching authorities over **mutual TLS**. In production they're **VNet-integrated**.

# How the network fits together — simply.

## One entry, one experience — private and sovereign underneath.

- **One public entry:** a single **Azure Front Door** (web firewall) and one UI codebase — the citizen always sees the same portal.
- **Sovereign by design:** each country has its *own* private front-end origin — a **Static Web App reached through a private endpoint** in that country's `web` subnet, with no public origin.
- That `web` subnet in each spoke is the **private door** to the front-end, not a separate copy of the app.
- App, data, AI, Logic Apps are private and in-country too; a **shared hub** handles firewall, DNS, cross-border and monitoring.



# Design patterns we use.

## Agents-as-Tools

On a phone call, the speech model decides by itself when to call a function or pass to a human.

## Gateway + Saga

One API gateway per country. The cross-border case is a 6-step workflow that rolls back if a step fails.

## System of record — Dataverse

Dataverse holds each case — surfaced via Power Apps today, D365 Customer Service tomorrow. Writes through workflows, reads through the gateway with a cache.

## Strangler fig

The caseworker app writes to a generic Dataverse table today, the canonical D365 Customer Service case entity tomorrow — one switch.

## Sidecar

The voice service pairs a call handler with a real-time audio bridge.

## Circuit breaker

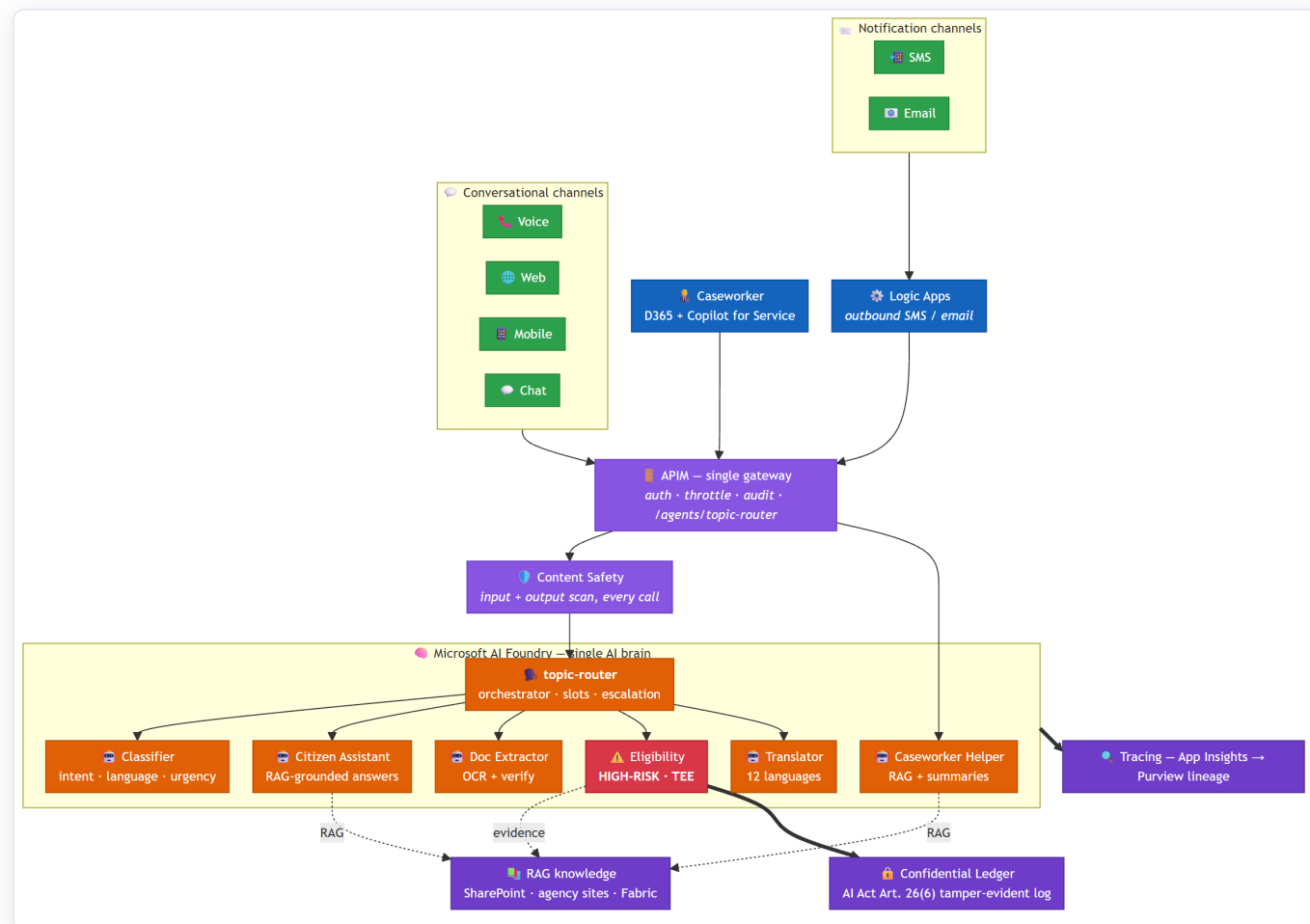
A failing national endpoint switches to a human queue — citizens never see a timeout.

# The AI Brain.

One brain on Azure AI Foundry · one orchestrator · six specialists · safe by design

02

# The AI Brain — every channel, one gateway.



**One brain, not seven chatbots** — every model call goes through Azure AI Foundry, routed by the **Topic Router**.



# How a citizen's question flows through the brain.

## 01 The channel asks the brain

Web, mobile and phone reach the Topic Router through the one private gateway — never a model directly.

## 02 The router understands and remembers

It finds the intent and the language across 12 languages, and keeps the short conversation for 24 hours.

## 03 It passes the task to a specialist

Classifier, Translator, Document Reader, Citizen Assistant, Caseworker Helper — or the high-risk Eligibility agent.

## 04 The specialist does one job well

The Assistant answers only from cited sources; the Reader extracts a payslip; Eligibility proposes a result with evidence.

## 05 The answer returns — traced and sourced

Every reply carries a trace number and its sources. On voice, the model decides when to call a tool or a human.

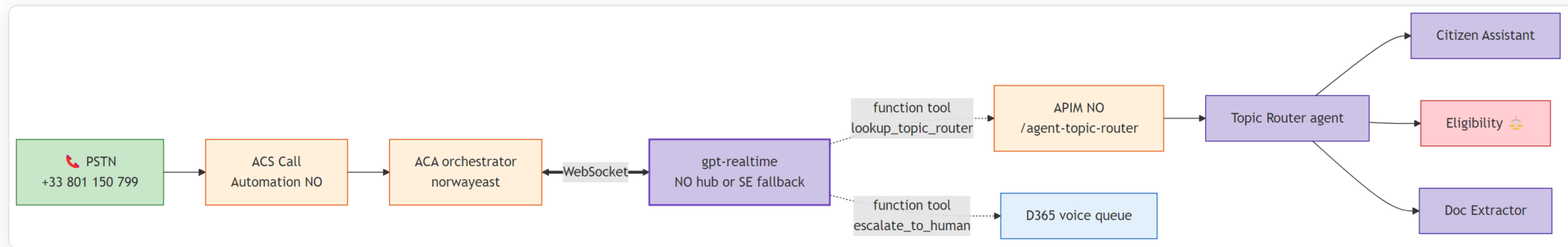
| Six specialist agents, two function tools, one orchestrator — seven agents in all, always under one human caseworker.

# Agent catalogue.

| Agent                                                                                                         | Purpose                                                         | Model                   | EU AI Act class       |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------|-----------------------|
| Topic Router                                                                                                  | Orchestration · 12 languages · slot-filling                     | Fast                    | Limited               |
| Request Classifier                                                                                            | Intent · agency · language · urgency                            | Fast                    | Limited               |
| Translator                                                                                                    | 12 languages · keeps admin terms exact                          | Strong + Translator     | Limited               |
| Eligibility Pre-Assessor<br> | Result + evidence · runs in a protected enclave · ledger-logged | Strong + rules          | High (public service) |
| Citizen Assistant                                                                                             | Answers only from the cited public knowledge base               | Strong, grounded        | Limited               |
| Document Reader                                                                                               | Passport / payslip / lease extraction                           | Fast + Doc Intelligence | Limited               |
| Caseworker Helper                                                                                             | Summarise · draft replies · suggest next action                 | Strong, grounded        | Limited               |

**No agent makes a final decision.** Every result is a *proposal* to a human caseworker, who confirms, changes or rejects it.

# Voice channel — the model uses tools on its own.



Citizen dials → the call lands on Azure Communication Services → a small service opens a two-way audio stream to the speech model.

**The model decides on its own** to call `lookup_topic_router` or `escalate_to_human` (a warm transfer). This is the Microsoft Agent Framework *Agents-as-Tools* pattern, on a real phone call.

# Eligibility model lifecycle.

## **01** Champion-challenger

A new version runs in shadow on 5% of traffic for one week, then we evaluate it.

## **02** Parity gate per language

A gold test runs in all 12 languages. If one language is too weak, the release is blocked.

## **03** Drift detection

A daily test on inputs and outputs. Drift opens a security incident and forces a new evaluation.

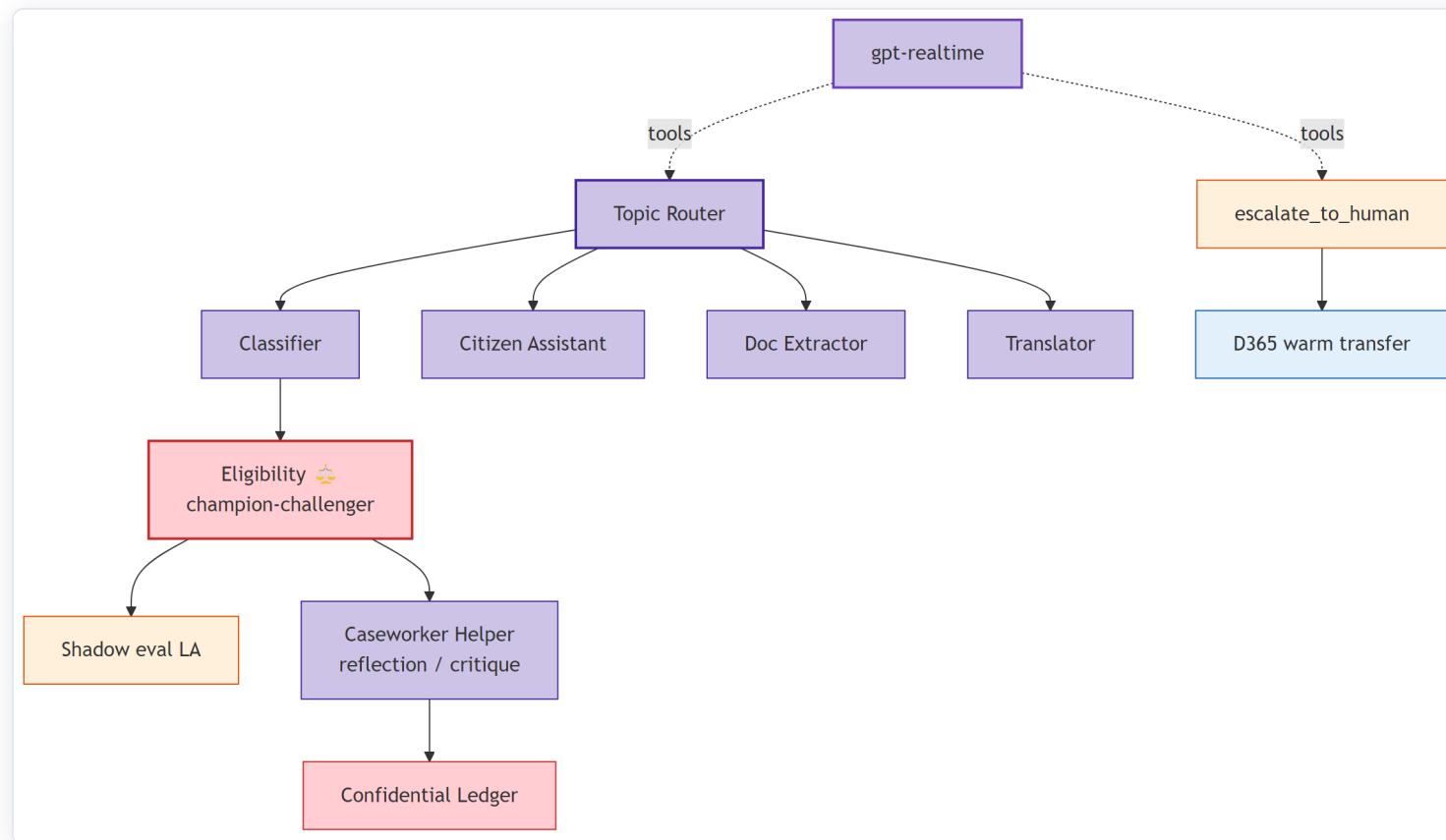
## **04** Bias monitoring

Fairness tests across age, location and channel for 30 days. A problem routes cases to a senior reviewer.

## **05** Rollback in seconds

Versions never change. Switching the alias restores the previous one at once, with an audit record.

# Multi-agent coordination — more than a chatbot.



**Handoff** (router → six experts) · **Reflection** (the helper reviews the eligibility result) · **State graph** (the case is a 6-state workflow that can undo a step) · **Function tool** and **warm transfer** on voice · **Shadow / canary** for every model change.

# Trust built into the brain.

## Content Safety

A jailbreak and prompt-injection detector runs on every turn; a block raises a security event.

## Evaluations

Quality is tested on every change — language parity, grounding and safety.

## AI Act register

Each agent is declared with its purpose, risk level and limits, in a versioned file.

## Confidential Ledger

Each high-risk result is hashed into a log that cannot be changed.

## Human-in-the-loop

No agent decides. A caseworker confirms, changes or rejects every result.

## Transparency

The citizen is always told AI is used — a chat badge and a spoken message in 12 languages.

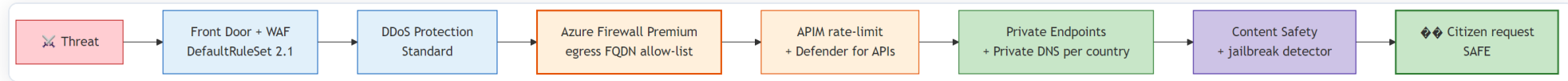
The Eligibility agent is the only high-risk part. It runs inside a **protected enclave** — its data is encrypted in memory, even from an administrator — and it never decides alone.

# Security & compliance.

How we made it provable — eight EU laws mapped to controls in code

03

# Defence in depth — many layers.



## Edge — application

Azure Front Door and web firewall — common-attack rules, bot protection, rate limits

## Edge — network

Standard DDoS protection on every network

## Traffic out

Azure Firewall · destination allow-lists · TLS inspection



# Identity & data sovereignty.

## IDENTITY

- One citizen-identity system per country (Microsoft Entra External ID)
- National eID via a certified broker — MitID · BankID + Freja+ · ID-porten
- Support for the EU Digital Identity Wallet
- Admins get time-limited access only, through one jump-host

One encryption key set per country · modern TLS in transit · two-way TLS to partners · field-level encryption for national IDs · public network access **turned off** everywhere.

## DATA — FIVE ZONES, ALL IN-COUNTRY

- **Live** — managed database and cache
- **Documents** — write-once, cannot be changed
- **Conversations** — kept for AI Act records
- **Knowledge** — search, locked per citizen and country
- **Analytics** — sovereign EU Fabric capacity

# How we made it compliant — the story.

## 01 We started from the law, not the features

Eight bodies of EU law touch one interaction — a single voice call is data protection, accessibility, e-identity, cybersecurity and AI law, all at once.

## 02 Each obligation became a control in code

Every rule maps to a real platform mechanism, with an evidence file behind it.

## 03 We made the evidence provable

One trace number follows every interaction, from the browser to the model and back, kept for two years.

GDPR — data protection

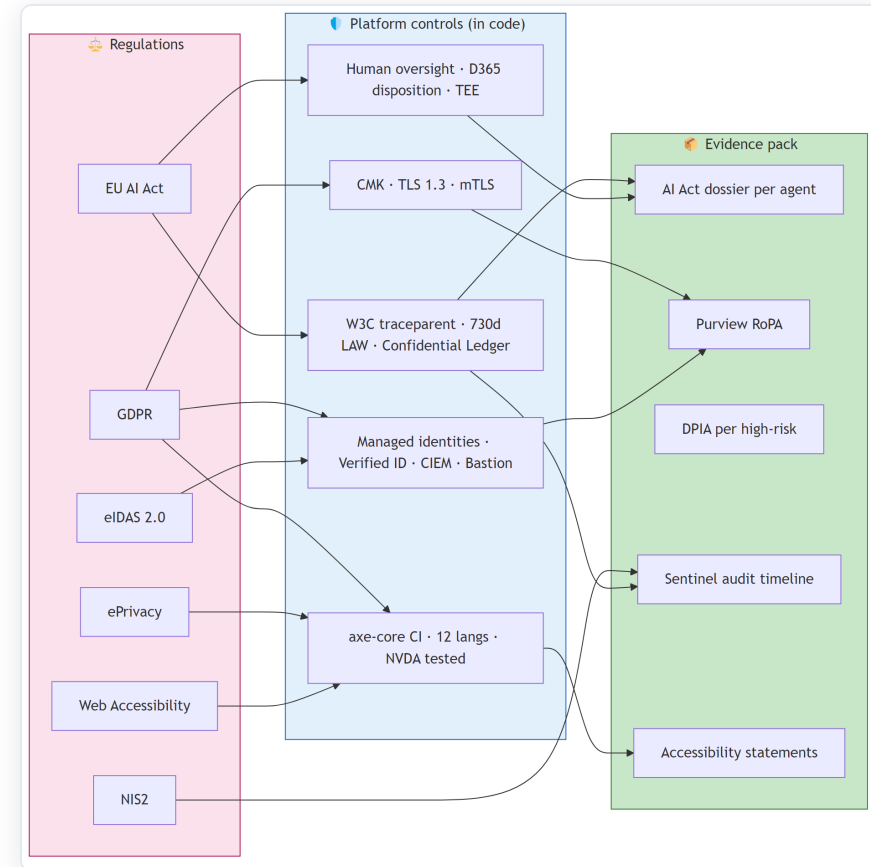
EU AI Act — high-risk AI

eIDAS — e-identity

NIS2 — cybersecurity

ePrivacy — cookies

WCAG — accessibility



# EU AI Act — article by article.

| Article         | Requirement                                     | UDCSP delivery                                     |
|-----------------|-------------------------------------------------|----------------------------------------------------|
| Art. 12         | Automatic records ≥ 6 months for high-risk      | Model logs kept <b>730 days</b> = 2× the minimum   |
| Art. 13         | Transparency for deployers                      | A register entry per agent · tests per language    |
| Art. 14         | Human oversight                                 | A caseworker confirms every result · ledger-logged |
| Annex III §5(b) | High-risk = access to essential public services | The Eligibility agent is declared high-risk        |
| Art. 50         | The citizen must know AI is used                | AI badge on chat · spoken message in 12 languages  |

The AI Act, GDPR, ePrivacy, eIDAS, NIS2 and accessibility law are each mapped to platform controls. The full mapping is in `docs/biz/datacompliance.md`.

# Security in action — a prompt injection, stopped.

A message that tries to leak the hidden instructions, or change a result, is caught at **three independent layers** — three different Azure tools:

## 01 Azure API Management

The gateway sees the strange token rate and blocks the data leak.

## 02 Azure AI Content Safety

The jailbreak detector raises a security event and opens an incident.

## 03 Fixed eligibility rule

A deterministic plug-in rejects the request before any model runs.

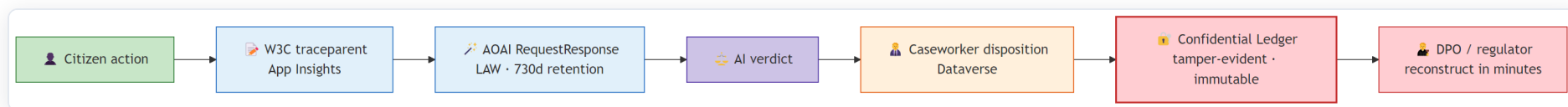
## THE AUTOMATIC RESPONSE

- The session is isolated at once
- The citizen flow is brought back safely
- An audit pack is saved for review

0

citizen data exposed

# Compliance in action — replay a decision, months later.



Hans, a data-protection officer, takes the citizen's **trace number** and rebuilds the whole decision — the model, the tokens, the result, the human choice, and the cryptographic ledger proof.

The AI Act minimum is six months. We keep **two years**.  
From the request to a full audit pack: **under 10 minutes**.

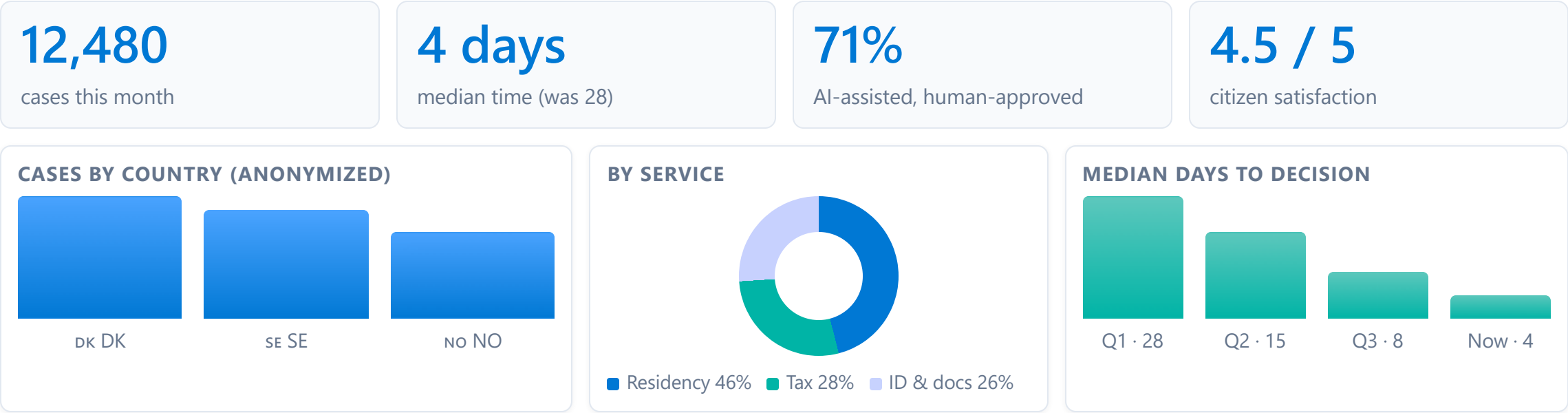
# Monitoring & FinOps.

One anonymized central view · end-to-end tracing · cost per token

04

# One central view — anonymized, in Fabric & Power BI.

Leadership needs the big picture. But raw citizen data never leaves its country. So only **anonymized, aggregated numbers** travel to a central view.



Only anonymized, aggregated numbers leave each country — never a name, never a case file. Built on Microsoft Fabric and Power BI · refreshed every hour.

# FinOps — cost under control.

## ALLOCATION & SHOWBACK

- Every resource tagged `country` , `workload` , `cost-center`
- Cost views per country and per workload
- Each ministry sees its own country's cost

## TOKEN BUDGET & CAPACITY

- A monthly token budget per agent, written in code
- The build **fails** if the total is over the pool
- Reserved capacity for steady models · pay-as-you-go for peaks
- Alert on a +30% jump in one day

| We treat AI tokens like any other budget line: planned, tagged, and checked at build time.



# Performance & reliability.

Fast for the citizen, always available — measured targets, automatic failover, chaos drills

05

# Performance & reliability.

**99.9%**

citizen web target, per country

**≤ 2 s**

voice answer, 95% of calls

**15 min**

recovery point — max data loss

**4 h**

recovery time to the backup EU region

## Active-passive per country

Azure Front Door sends traffic to the healthy region. Failover in under 5 minutes.

## Backup & disaster recovery

Database, cache, storage and agent hosts. One recovery vault per country.

## Chaos drills

We break things on purpose — region failover, network cut, database failover. Monthly in test, quarterly in production.

## Autoscale & caching

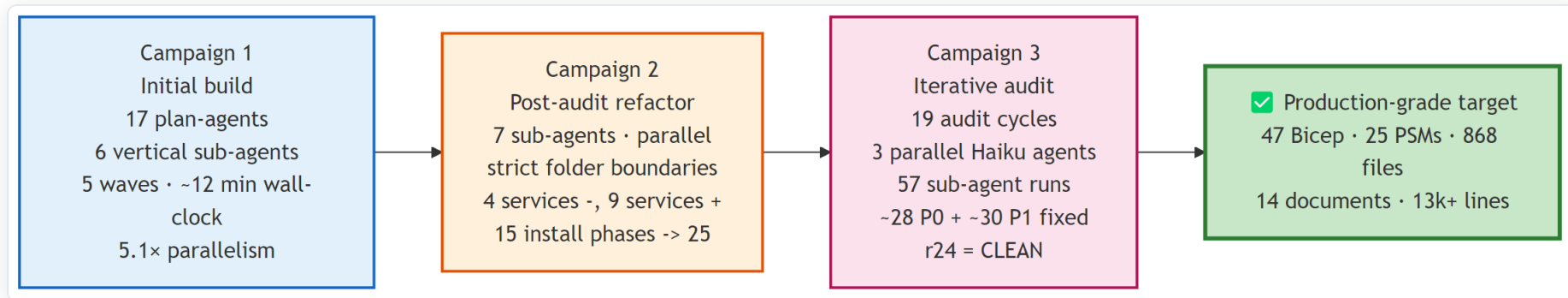
Services scale out under load. Reads are cached at the gateway, so peaks stay fast.

# How we built it.

The platform itself was built by an agent swarm — in under 45 minutes of wall-clock time

06

# UDCSP was built by an agent swarm.



3

multi-agent campaigns

~5×

faster than sequential

24

audit rounds to first clean

868

tracked files produced

Three campaigns — build, refactor, then repeated audits — produced the platform. Six sub-agents owned separate folders and ran in parallel. Each round produced fix commits until round 24 was the first fully **clean** one. The one-command installer is the executable summary of it all.

# Everything is in the open — one repository.

## THE STORY STARTS IN THE DOCS

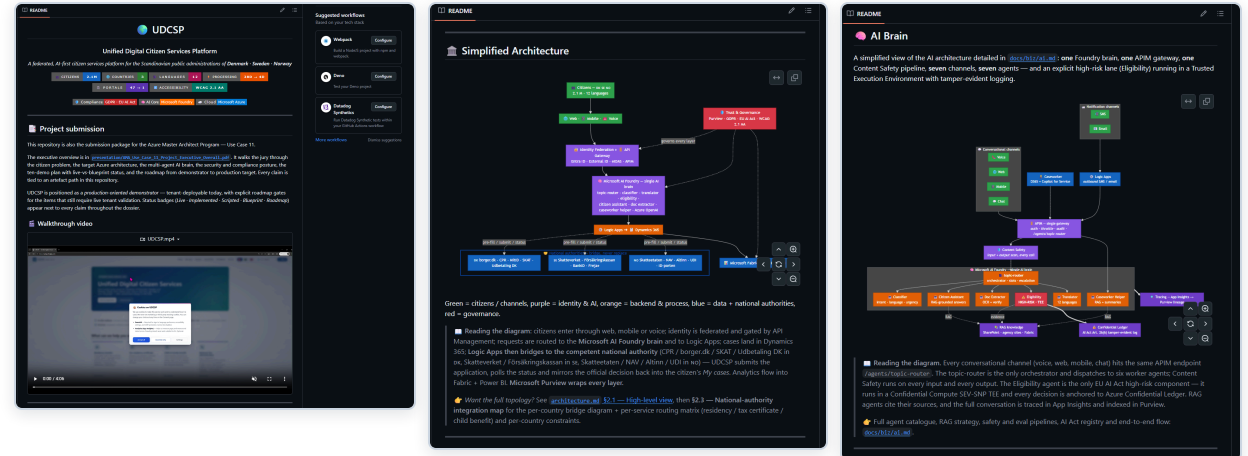
- `/docs/biz` — the business story: who we build for, every channel, data-and-compliance, demo scenarios, and an acceptance recipe for "done".
- `/docs/tech` — the engineering: architecture deep-dive, data model, network design, one-command install guide, disaster-recovery runbook.
- `/infra` — the Bicep that builds every zone ·  
`/scripts` — the installer.

SHORT LINK

[aka.ms/UDCSP](https://aka.ms/UDCSP)

GITHUB

[github.com/fredgis/UDCSP](https://github.com/fredgis/UDCSP)



# One idempotent deploy — provision.

One script, one click — and you can run it again safely.

- Checks prerequisites, signs in to Azure, registers the **Entra ID** identity app.
- Installs, **builds**, and runs the **tests** before anything is deployed.
- Provisions all the infrastructure **as code** — Bicep, through `azd`.
- **Idempotent**: a second run reuses what exists — never a duplicate.

```
Fabric Storyboard Copilot - One-Click Deploy

▶ Checking prerequisites
✓ Node.js v24.14.0
✓ npm v11.9.0
✓ Azure CLI v
✓ Azure Developer CLI (azd)

▶ Authenticating with Azure
✓ Signed in as
✓ Subscription:
Authenticating azd (fresh login)...
✓ azd authenticated

▶ Gathering configuration
Environment name (e.g., dev, staging, prod) [dev]:
✓ Environment: dev
Azure region [eastus2]:
✓ Location: eastus2
Deploy a NEW Azure OpenAI resource? (y/n) [n]:
Existing Azure OpenAI endpoint URL: https://.openai.azure.com/openai/v1/
✓ OpenAI endpoint: https://-openai.azure.com/openai/v1/
✓ OpenAI deployment: gpt-4o

▶ Configuring Entra ID
Create a NEW Entra ID app registration? (y/n) [y]:
Registering app: Fabric Storyboard Copilot (dev)
✓ App registered:
Setting Application ID URI...
Exposing access_as_user scope...
✓ Scope: api:// /access_as_user
Adding redirect URIs...
Adding API permissions...
Authorizing Office clients...
✓ Office clients authorized (SSO)
Generating client secret...
✓ Client secret created (valid 1 year)
Creating service principal...
Granting admin consent...
✓ Entra ID setup complete

▶ Installing dependencies
✓ Frontend packages installed
✓ Backend packages installed

▶ Building application
✓ Frontend → dist/
✓ Backend → api/dist/

▶ Running tests
✓ Frontend tests passed
✓ Backend tests passed

▶ Provisioning Azure infrastructure (Bicep via azd)
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13

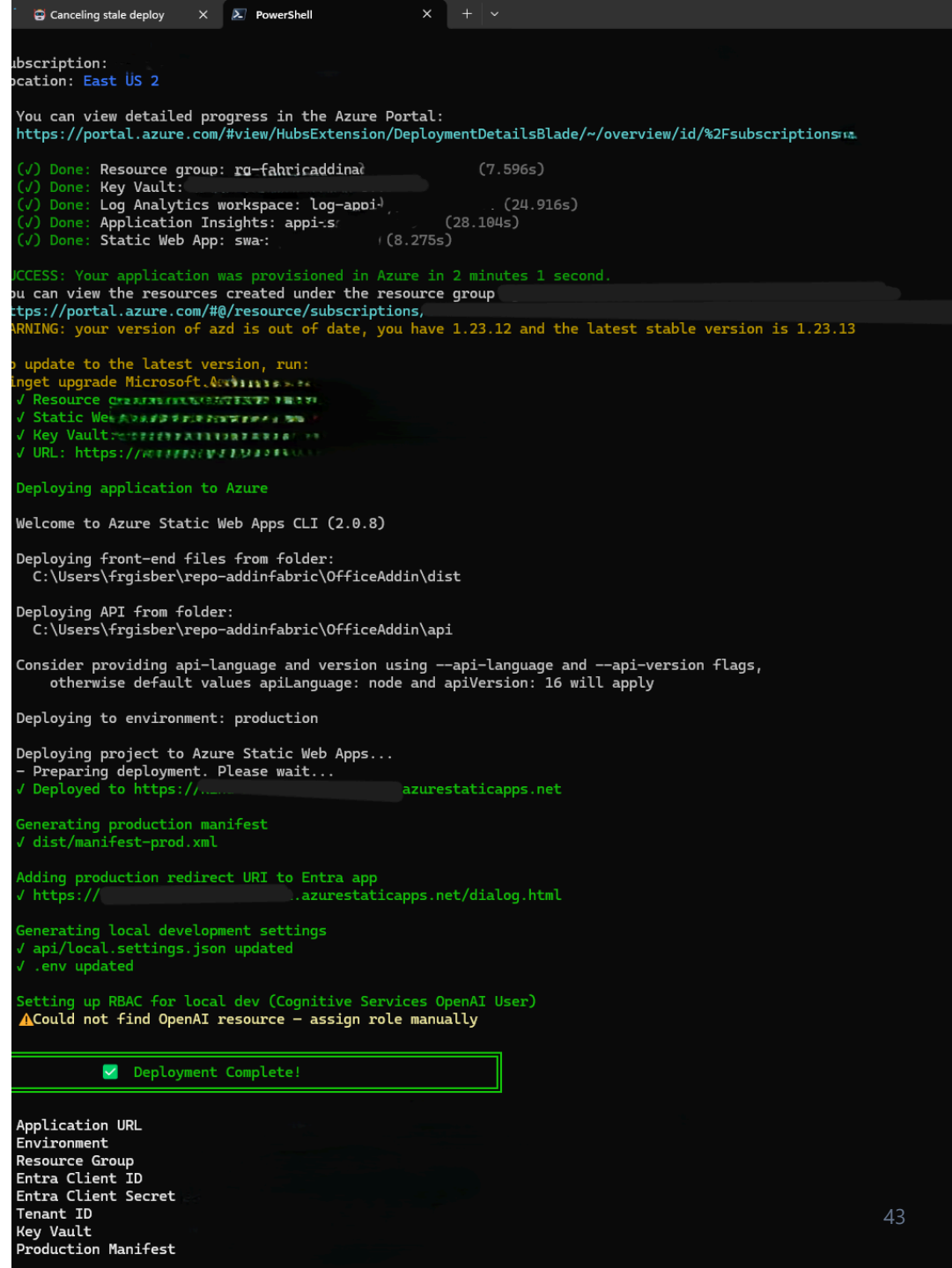
To update to the latest version, run:
winget upgrade Microsoft.Azd
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13

To update to the latest version, run:
winget upgrade Microsoft.Azd
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13
```

# One idempotent deploy — release and verify.

**It deploys, verifies, and reports — every time the same way.**

- Provisioned in **~2 minutes**, then the app is deployed to Azure.
- Generates the **production manifest**, wires redirect URIs and **RBAC** (role-based access control).
- Ends with **Deployment Complete** and a summary of every resource and URL.
- The same script runs in the **pipeline on every change** — repeatable, not hand-made.



# Real scenario.

Life imitates the demo — a story we did not plan

07



# "I bought my car back."



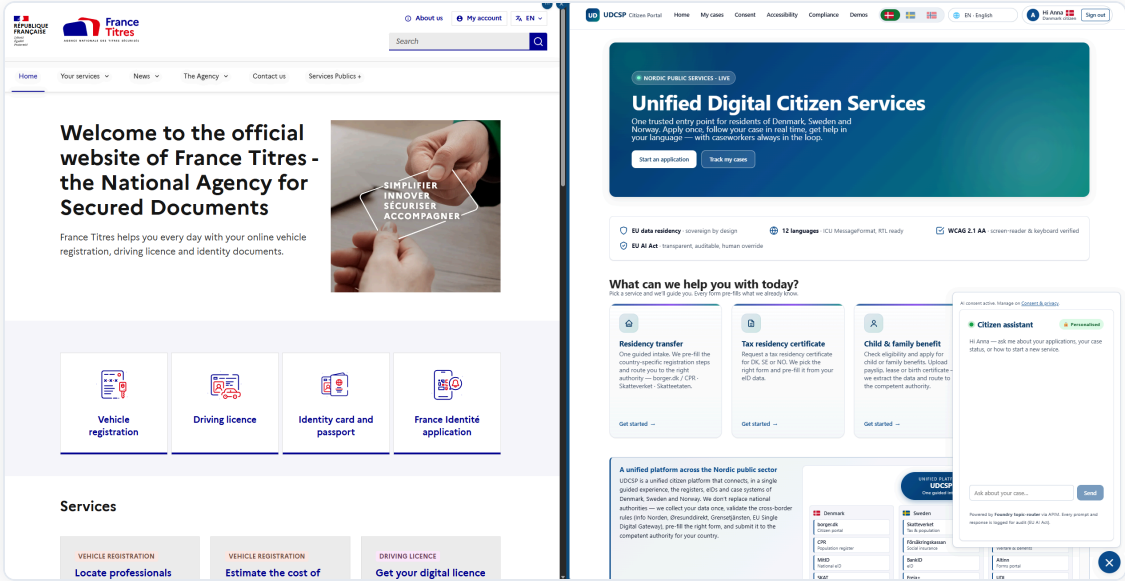
To change my licence plate, I signed in to the **official French government portal** — France Titres.

I had **never seen it** while building UDCSP.

And there it was — the same one-front-door layout, the same service cards (vehicle registration, driving licence, ID & passport), the same assistant in the corner.

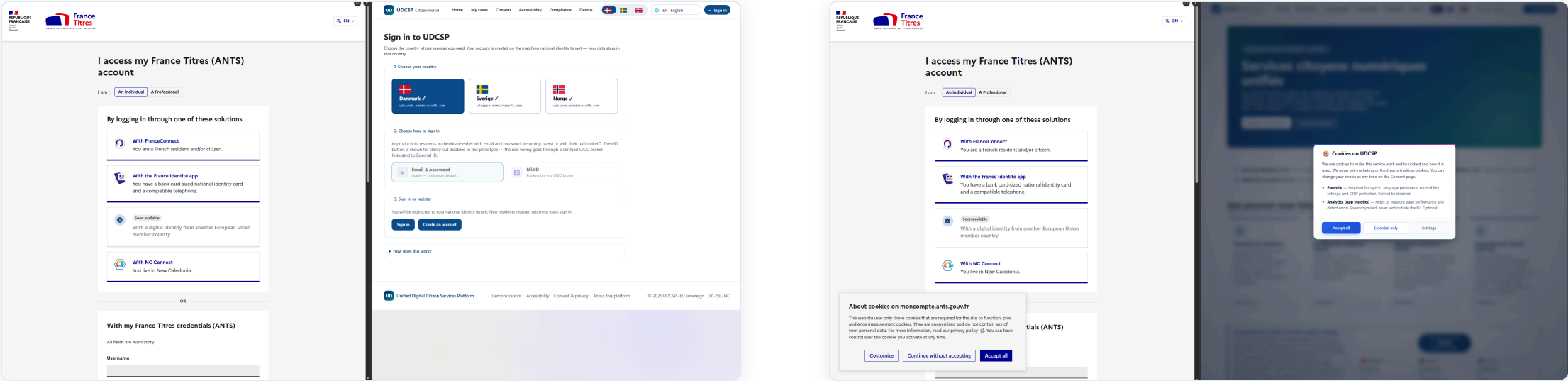


FR FRANCE TITRES — THE REAL THING



UDCSP — OUR DESIGN

...and we'd built it first.



Pick your country, sign in with a national eID, and — "coming soon" — an **EU cross-border identity**. That is our model, almost line for line. **We designed it before we ever saw it.**

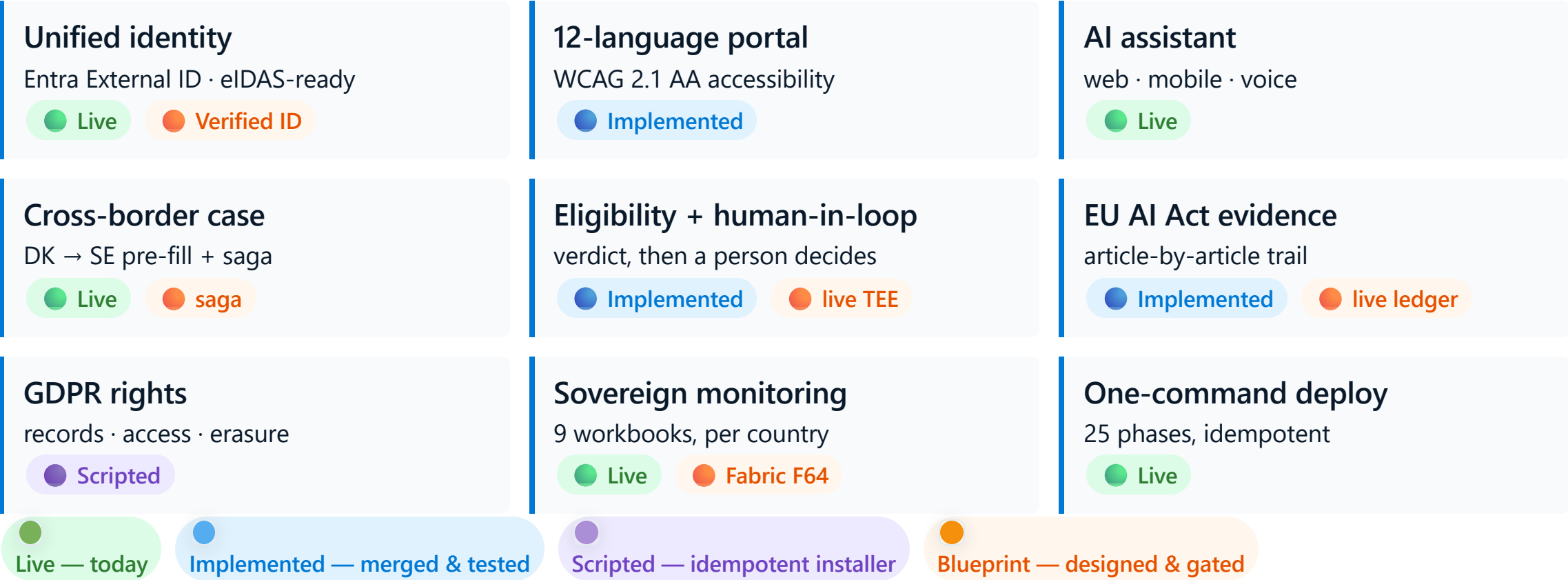


# Status & roadmap.

What runs today · what's blueprint · how it reaches production

08

# Requirement coverage — the honest scorecard.



The working core carries the demo; each blueprint brick has a named gate.

# Four gates from accelerator to production.

## GATE 1 · WK 1–2

### Tenant validation

Confirm model deployments, Fabric F64 capacity, DDoS plan & Foundry hub.

## GATE 2 · WK 3–6

### Live confidential compute

Eligibility on a SEV-SNP enclave + Confidential Ledger.  
**Demo 6 → Live.**

## GATE 3 · WK 6–14

### Partner-agency integration

Mutual-TLS agreements + production federation hub.  
**Demo 1 → real authority.**

## GATE 4 · WK 14–20

### D365 + Copilot for Service

Strangler-fig repoint of case writes. **Demo 5 → Live.**

Today → Gate 1 → Gate 2 → Gate 3 → Gate 4 → production. **Each gate is independently shippable** — every blueprint brick has an owner, a window, and a demo it flips to live.

# Demo.

10 minutes, live on a real tenant

09

# The run of show.

## 01 Anna moves DK → SE ● Live

Sign in with a Danish eID · upload a passport and lease · the AI reads, translates and proposes a result · consent · the case crosses the border.

## 02 Lars calls the voice line ★ ● Live

A blind citizen dials a real free number · the model answers in Norwegian and routes itself · a human transfer is offered.

## 03 Maria uses a screen reader in Polish ● Live

The whole portal in Polish, keyboard only, screen-reader clean — accessibility as a citizen right.

## 04 Erik photographs a payslip on iPhone ● Live

The same web app on mobile · native iOS picker · structured fields and a result, inline.

## 05 Ole builds it from a clean tenant ● Live

One command: 25 phases, sample data seeded, smoke tests green.

Hero moment — **a real phone call**: dial **+33 801 150 799**, hear the model answer, and watch the live dashboard update within two minutes.



# One command, from clean tenant to running platform.

```
git clone https://github.com/fredgis/UDCSP
cd UDCSP
pwsh ./scripts/install/Install-UDCSP.ps1 `
    -Environment dev -Zone all `
    -SeedSyntheticData -Verbose
```

- 25 phases in order: landing zone → identity → security → data → monitoring → gateway → Foundry → Dataverse → frontend → voice → governance → quality
- Sample data is seeded in parallel with the frontend
- A smoke test runs at the end (identity · gateway · AI · case creation · dashboard · accessibility)
- An HTML report is saved in `scripts/install/reports/<timestamp>/`

From `git clone` to a working federated platform with real sample data: **one command**. The same script runs in our pipeline on every pull request.



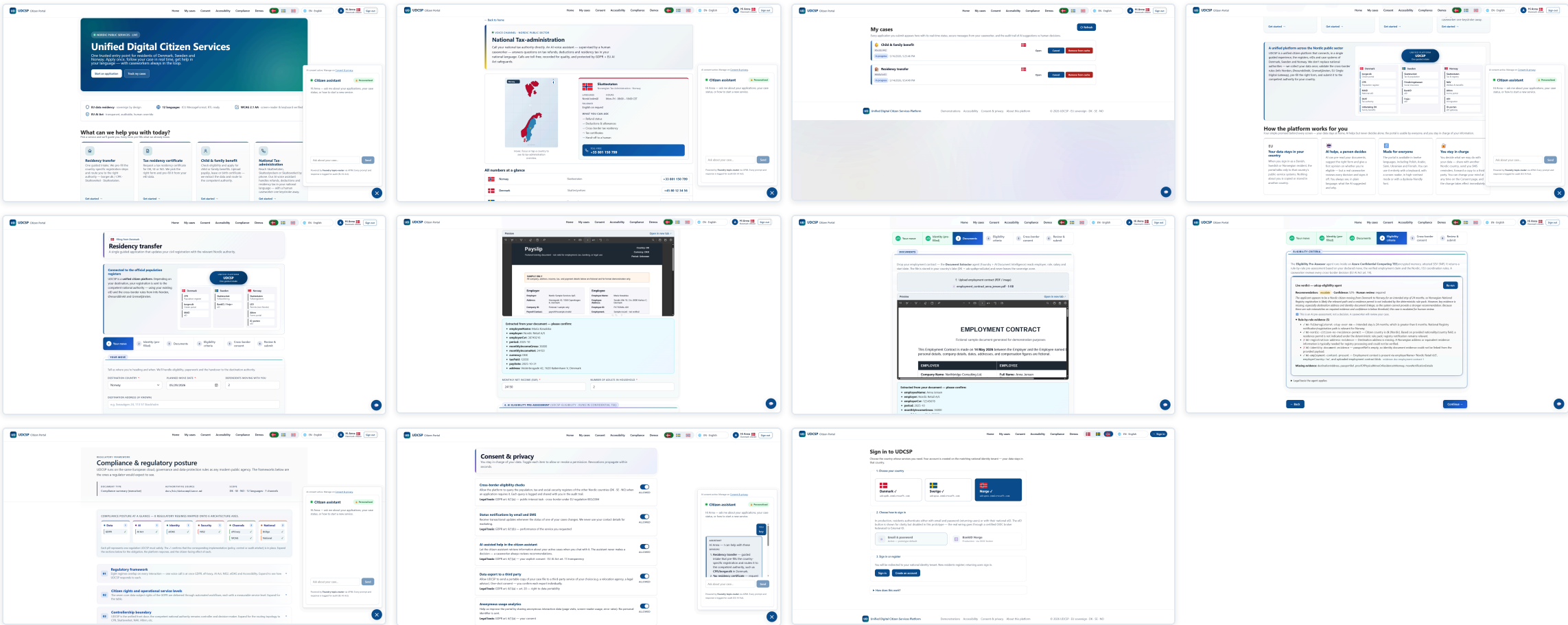
CLOSING

# UDCSP — a production accelerator with a named path to production.

3 sovereign zones · 7 AI agents · one private path from the portal to the case · every decision anchored to a regulation · every claim labelled and provable on a real Azure tenant. A working core you deploy with one command today — and a four-gate roadmap that takes the blueprint bricks to production. A real French government portal, seen by accident, proved the architecture was right.

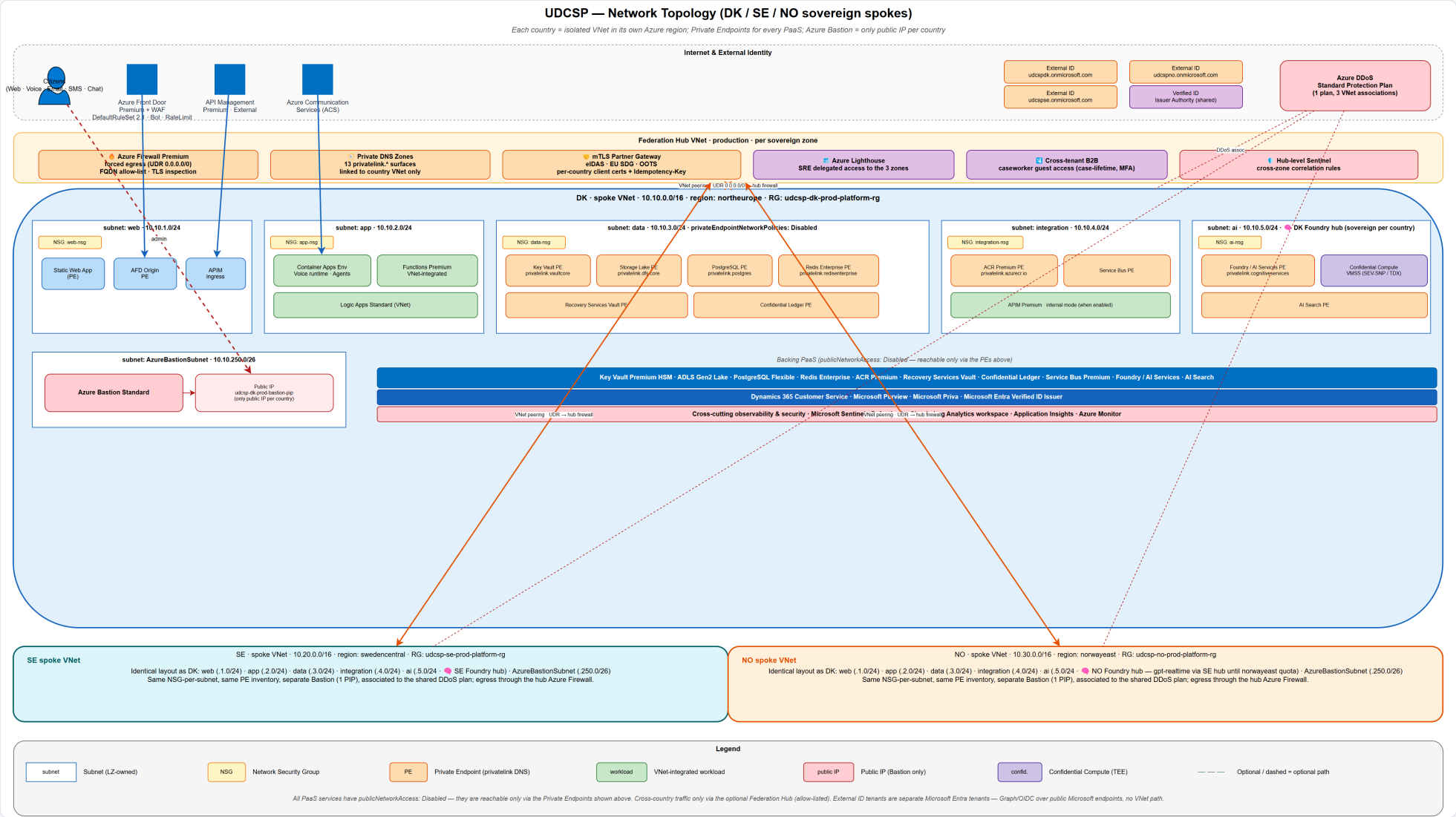
Frederic Gisbert · June 2026 · [github.com/fredgis/UDCSP](https://github.com/fredgis/UDCSP)

# Annex — the platform, screen by screen.

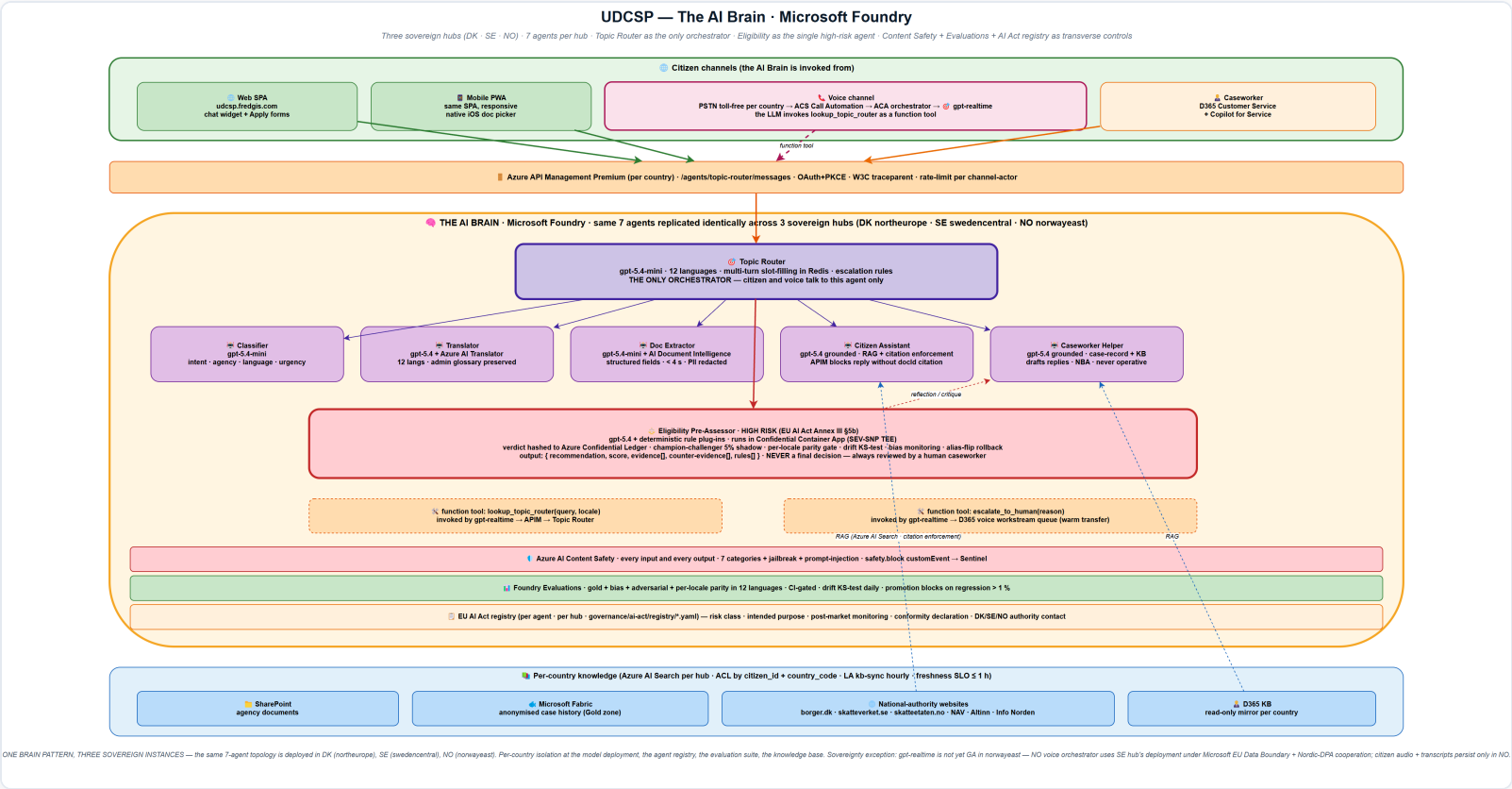


1 Home · 2 Contact / voice · 3 My cases · 4 Assistant · 5 Residency intake · 6–7 Document reading · 8 Cross-border result · 9 Compliance · 10 Consent · 11 Sign-in. One web app, twelve languages, three channels — the same React and TypeScript code on desktop, tablet and

# Annex — network topology in full.



# Annex — the AI Brain blueprint in full.



One orchestrator and six specialists on Azure AI Foundry, the high-risk eligibility agent in a sealed enclave, and the EU AI Act register wired into every model call — the dense source map behind the simplified gateway slide.

# Annex — telemetry pillars in full.

METRICS / LOGS

Azure Monitor

3 workspaces, one per country · 180 days hot · 7 years cold · never joined

DISTRIBUTED TRACE

Application Insights

One trace number end-to-end · front door → gateway → workflow → Dataverse → agent → model

AI QUALITY

Foundry Evaluations

Continuous · drift monitors · language parity · bias monitoring

AI ACT EVIDENCE

Confidential Ledger

Records that cannot be changed · joins logs, AI and Dataverse in one query

ACTIVE

Synthetic + real-user

5 external regions · 60-second probes · load times per language

DASHBOARDS

9 workbooks

platform health · citizen-journey funnel · AI-decision traces · per country